



SUDHARSAN RAGOTHAMAN

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EDUCATION

Master of Science, Engineering Data Science

University of Houston, Houston, Texas, USA

Aug 2021 – Dec 2022

GPA: 3.70/4

Courses/Research Interests: Computer Vision, Machine Learning Models, Probability and Statistics for ML, Deep Learning

Bachelor of Technology, Mechanical Engineering

National Institute of Technology, Trichy

Jul 2017 – May 2021

GPA: 3.2/4

Courses: Data Analytics, Computational Techniques, Optimization Techniques, Introduction to C++, CS with Python

SKILLS

Programming Languages: Python, C++, MATLAB, HTML, XML, JavaScript

Software & Tools: AWS, Jupyter Notebook, PyCharm, VSCode, Unix, Git, Excel, PowerBI, IICS

Machine Learning Libraries: NumPy, Scikit-learn, Pandas, PyTorch, OpenCV, SciPy, Matplotlib, boto3, TensorFlow, NLP

Databases and Bigdata Technologies: Snowflake, Spark, Databricks, HDFS, MySQL, Kafka Streaming, Airflow, Tableau

Cloud services: AWS S3, EC2, AWS EMR, Sagemaker, AWS Glue

PROFESSIONAL EXPERIENCE

Data Engineer , Javen Technologies, Minneapolis, MN-Client-Solventum

Jul 2024 – Present

- Designed and optimized **Snowflake data marts** using **dimensional modeling**, **clustering keys**, and **materialized views**, enabling high-performance querying across OTC, Commercial Ops, and MDM datasets.
- Built **incremental ELT pipelines** using **Snowflake Tasks + Streams**, implementing change data capture and dependency-driven processing for staging, transformation, and upsert layers.
- Developed **complex SQL queries**, **stored procedures**, and **dynamic SQL frameworks** to perform investigative data analysis and automate metadata-driven transformations.
- Utilized **Time Travel** and **zero-copy cloning** for safe data recovery, point-in-time comparisons, regression testing, and parallel QA environments with zero impact on production data.
- Performed **large-scale query optimization and data profiling** across multi-million record datasets to identify anomalies, improve performance, and ensure data reliability.
- Delivered real-time **Power BI dashboards** via **Snowflake DirectQuery**, with DAX-based KPIs, row-level security, and semantic model optimization for fast and secure stakeholder access.
- Architected a multi-agent orchestration proof-of-concept on **Azure AI Foundry**, using the Microsoft Agent Framework (**MAF**) to coordinate specialist agents via AIProjectClient, demonstrating scalable orchestration patterns applicable to enterprise data workflows.
- Built a monitoring and validation framework for a Microsoft Fabric Data Agent, combining a golden-question test suite, an LLM-as-judge evaluation layer, and automated freshness checks with Teams Adaptive Card alerting to ensure AI-driven data responses stayed accurate and current.
- Employed **GitHub for version control** and **Jira for sprint/project management**, integrating CI/CD practices and environment-specific branching strategies across IICS, Snowflake, and Power BI artifacts.

Data Scientist, Trovadatum Inc, Georgia, USA

Feb 2023 -Jul 2024

- Conducted **exploratory data analysis (EDA)** and large-scale data manipulation using Python (Pandas, NumPy) and SQL across operational and analytics databases.
- Built an resume bot, a GENAI bot using LLM and Langchain models in vision of creating and interactive Resume.
- Created end-to-end LLM project and completed thorough research on different LLM models like llama2chat, OpenAI in building an interactive customer service care bot for the client.
- Built **interactive Jupyter notebooks** to analyze model performance, data distributions, and feature behavior.
- Developed Apache Spark code snippets on Databricks as a part of data exploration with data streaming from snowflake using Kafka streaming and workflows are created using Airflow to automate the pipeline.

RESEARCH EXPERIENCE

Cloud-Controlled Drone Computer Vision Model

- Utilized AWS SageMaker to develop a state-of-the-art ML model for object detection during drone operations. The trained model was stored in the S3 bucket, leading to increased accuracy by 40% and efficiency in identifying objects.
- Implemented Lambda functions and API Gateway to automate object detection inferences on camera images stored in the S3 bucket, reducing the need for manual intervention by 90%, and streamlining processes.

Deep Learning Approach for Land Cover Classification Using Satellite Imagery- CVML

- Employed deep learning to create a powerful multi-class image segmentation model for precise land cover prediction in environmental monitoring, improved segmentation accuracy by 90%.Led U-Net and DeepLabV3+ implementation to address dataset class imbalance. Experimented with loss functions, including cross-entropy, dice, and Tversky.

Audio and Text Sentiment Analysis with Speech Emotion Classification

- Utilized Google API to convert audio segments into text, subsequently conducting sentiment analysis on a speech emotion dataset. Employed a diverse set of supervised models including Naïve Bayes, Decision Tree to evaluate the emotional context within audio segments. Achieved an impressive overall accuracy rate of 91.9% through the development of a Hybrid model.

INTERESTS & ACHIEVEMENTS

- Overall bronze medalist in the tournament conducted by World Taekwondo Federation of India.
- Snowflake **SnowPro Core Certified**.
- Python **PCEP Certified**